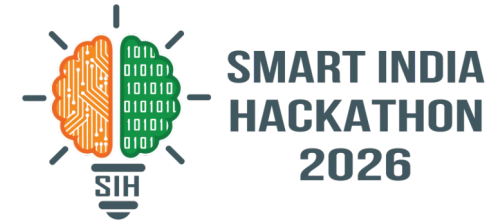


SMART INDIA HACKATHON 2026



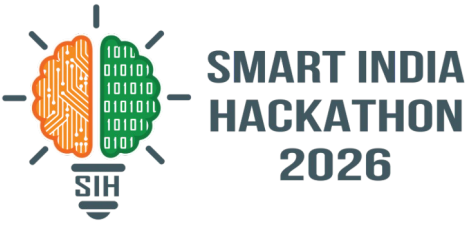
TITLE PAGE

PARAKH — a review queue, not a verdict

Anomaly flagging for MPLADS implementation records

- Problem Statement ID – SIH26102
- Problem Statement Title – Fraud & Anomaly Detection in MPLADS Scheme
- Theme – <<fill from the SIH portal>>
- PS Category – Software
- Team ID – <<fill from the SIH portal>>
- Team Name – <<registered team name>>





IDEA TITLE

PARAKH — a review queue for MPLADS implementation records. Not a verdict.

THE PROBLEM

- MPLADS funds ₹5 crore of local works per MP each year — 1,31,437 works this term
- Sanctions, payments and completion status sit in separate exports that are never joined
- No single place a citizen, auditor or ministry can see the scheme whole
- Manual audit reaches a small sample, years after the money moved
- An automated tool that names culprits is a defamation engine, not oversight

OUR SOLUTION

- A batch pipeline joins the exports into one queryable corpus
- 13 detectors surface works that warrant a second look
- Every flag states its evidence in one plain, checkable sentence
- Every flag carries a plausible innocent explanation, shown by default
- Reviewers close a flag with a note — suppressed, never deleted, always reversible

INNOVATION AND UNIQUENESS

- Flags attach to works. Members of Parliament are never ranked
- Two tiers, always visible: A is arithmetic, B is peer-relative
- Detectors abstain below 30 peers instead of guessing on noise
- Restraint enforced by tests in CI, not promised in a slide
- Ships 13 of 16 planned detectors — and says so inside the product

1,31,437

works ingested, 18th term

13

detectors: 5 deterministic, 8 statistical

5

independent validation methods

0

rankings of any Member of Parliament

Why this is hard to copy:

The restraint is executable. A test in the build greps the whole codebase for a fixed list of accusatory words, and the assistant refuses to rank members at all — so the guarantee survives contact with a demo, a judge, or a rushed reviewer.

Python · Polars · DuckDB · scikit-learn · LightGBM · NetworkX · FastAPI · Next.js + TypeScript — entirely open source



TIER A — DETERMINISTIC INTEGRITY (5)

A1 Ledger contradiction — payment exceeds its sanction

A2 Phantom completion — complete, almost nothing paid

A3 Duplicate record — same work ID twice in the export

A4 Orphan sanction — missing agency, state or category

A5 Unverified completion — no photo evidence on file

TIER B — PEER-RELATIVE STATISTICAL (8)

B1 Peer cost outlier B2 Cost-per-unit outlier

B3 Stalled vs category P95 B4 Threshold bunching

B5 Work splitting B6 Agency concentration

B7 Year-end bunching B8 Near-duplicate funding

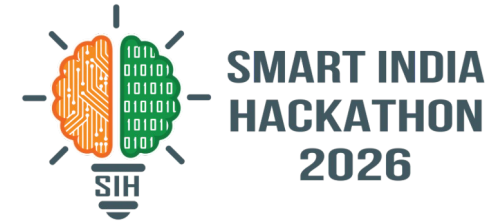
Always states its peer group and n; abstains below 30 peers

Reproducible by construction:

Raw CSV → Parquet → DuckDB is one command and a few seconds. The same input always yields the same flags, so any reviewer can re-run the pipeline and reproduce every number. 172 automated tests cover the detectors, the validation methods and the restraint rules themselves.



FEASIBILITY AND VIABILITY



Already running end to end on the full public corpus — 1,31,437 works, not a mock-up.

WHY IT IS FEASIBLE

- Runs today on the complete 18th-term corpus: 1,31,437 works, 1,09,006 payments
- Batch and deterministic — no GPU, no per-query model cost, no network at run time
- Open-source stack throughout; no licence, quota or vendor lock-in
- Rebuilds in seconds, so a reviewer can re-run it and reproduce every flag
- 172 automated tests, including the restraint guarantees

RISKS AND HOW WE HANDLE THEM

- Naming individuals would defame → flags attach to works; ranking members is refused in code
- Sparse source fields (quantity 4.9%, photos 23.8%) → detectors abstain rather than infer
- No labelled ground truth exists → five independent validation methods, results published in full
- Small peer groups mislead → Tier B abstains below 30 peers
- Benign patterns look irregular → every flag ships its innocent explanation alongside

Applied to ourselves:

We did not build the planned third tier — an Isolation Forest, a Local Outlier Factor model and a composite 0–100 priority score. 13 of 16 planned detectors shipped, and that is stated on the product's own Methodology page, not only in this deck.

FEASIBLE TODAY

Working software on real public data

SCALABLE NATIONWIDE

Same pipeline covers every state and both Houses

LOW COST TO RUN

Commodity hardware, open source, offline

From 1,31,437 scattered records to a queue a human can actually work through.

MoSPI AND DISTRICTS

A worklist ordered by evidence rather than by hunch or by complaint

AUDITORS

A reproducible shortlist with the arithmetic already shown and citable

CITIZENS AND PRESS

The same public data, finally joined and searchable in one place

IMPLEMENTING AGENCIES

A chance to explain an ordinary pattern before anything escalates

BEFORE

- Sanctions, payments and completion never joined
- Oversight by small sample, years after the spending
- Irregularities surface as allegations about people
- No way to show a pattern was ordinary after all

AFTER

- One corpus: 1,31,437 works, 1,09,006 payments, 769 agencies
- 37,701 works flagged with evidence attached
- 60,339 examined by every detector and cleared
- Reviewers close flags with a note — reversible, and audited

GOVERNANCE

Scrutiny that is traceable, reproducible and contestable

ECONOMIC

Attention aimed at the works where evidence justifies it

SOCIAL

Public money made legible without accusing anyone

NATIONAL SCALE

36 states, 733 MPs, both Houses, one pipeline

We do not tell anyone who is at fault. We show which records do not add up, what would explain them, and let a human decide.



RESEARCH AND REFERENCES



Public data, published methods, and a repository anyone can run.

DATA AND GOVERNMENT SOURCES

- eSAKSHI — MPLADS citizen dashboard, mplads.mospi.gov.in (all works, payments and completion exports)
- MPLADS Guidelines, Ministry of Statistics and Programme Implementation
- Comptroller and Auditor General of India — performance audit reports on MPLADS
- Lok Sabha and Rajya Sabha member directories, sansad.in

METHODS AND LITERATURE

- Leys et al., robust outlier detection with the median absolute deviation
- Blondel et al., fast unfolding of communities in large networks (Louvain)
- Ke et al., LightGBM: a highly efficient gradient boosting decision tree
- Saito and Rehmsmeier, precision-recall over ROC on imbalanced data
- Benford-style bunching and threshold-avoidance in public procurement

TECHNOLOGY

- DuckDB — duckdb.org · Polars — pola.rs
- scikit-learn — scikit-learn.org · NetworkX — networkx.org
- FastAPI — fastapi.tiangolo.com · Next.js — nextjs.org
- LightGBM — lightgbm.readthedocs.io

Source code and reproducible build:

github.com/Kartik-Garg-13/mplad_sih — the full pipeline, the 13 detectors, the five validation methods and 172 tests. Clone it, run two commands, and every figure in this deck rebuilds from the published source data.